

Homework 5

due 15.11.2023, 14:00

Exercise 5.1 - Gram Schmidt Process (6 points)

Very often it is necessary to compute an orthogonal, normalized basis for a subspace spanned by the N -dimensional vectors $\vec{x}_0, \dots, \vec{x}_{m-1}$. For simplicity we assume that this subspace is m dimensional. The standard method to construct the basis $\vec{b}_0, \dots, \vec{b}_{m-1}$ is the Gram-Schmidt orthogonalization, which proceeds as follows

```
loop l=0...m-1
   $\vec{t} \leftarrow \vec{x}_l$ 
  loop k=0...l-1
     $\vec{t} \leftarrow \vec{t} - \vec{b}_k(\vec{b}_k \cdot \vec{t})$ 
  end
   $\vec{b}_l \leftarrow \vec{t}/\|\vec{t}\|$ 
end
```

Write a program that carries out the Gram Schmidt orthogonalization in parallel. All vectors should be split into pieces of length $\bar{n} = N/\text{nproc}$ and no arrays of length N should be used at any time. To test your program, let process 0 read 11 vectors from the files ($N = 65536$) `x_0.dat`, ..., `x_9.dat` and `y.dat` that you can find in `/work/korzec/LAB2/ex5`. Read $\bar{n}$ entries at a time and send them to the appropriate process. Construct an orthogonal normalized basis from $\vec{x}_0, \dots, \vec{x}_9$. Compute the dot products of the resulting basis vectors with $\vec{y}$ and hand in your result (10 numbers).

Exercise 5.2 - Strong Scaling (4 points)

Study the strong scaling of your program with 1, 2, 4, 8, 16, 32 and 64 processes. Plot the speedup vs the process count. Plot also the perfect scaling curve, as well as a prediction from Amdahl's law (try to find a parameter r that describes your data best).